

Tier 0 structural / control

n = 40 judged pairs per family

syntactic_scramble

1/2

word order in MOVE/SUBTASK

move to the

→ ∅

judge: meaning preserved

1.000

cross_task_swap

1/10

whole trace ← another task

∅

→ because it's the closest and most direct path

judge: meaning preserved

0.025

Tier 1 word-level semantic

n = 40 judged pairs per family

direction_flip

1/2

left/right, up/down in MOVE

left

→ right

judge: meaning preserved

0.200

gripper_flip

1/2

grasp/release in PLAN

grasp

→ release

judge: meaning preserved

0.050

verb_swap

1/3

action verb → unrelated verb

move

→ hold

judge: meaning preserved

0.575

negation

1/2

prepend to SUBTASK/MOVE

∅

→ do not

judge: meaning preserved

0.100

Tier 2 object / causal chain

n = 24–29 judged pairs per family

subject_swap

1/5

referent → visible distractor

flat stove.

→ white bowl.

judge: meaning preserved

0.000

location_swap

spatial phrase in PLAN/TASK

left

→ right

judge: meaning preserved

0.276

adversarial_plausible

1/5

referent → 2nd-most-visible

flat stove.

→ wooden two layer shelf.

judge: meaning preserved

0.000

Tier N calibration nulls

n = 40 judged pairs per family

paraphrase_null

1/6

synonym in MOVE/SUBTASK

move

→ shift

judge: meaning preserved

0.975

bbox_jitter_null

1/3

±1px on VISIBLE OBJECTS

[[106, 165], [146, 207]],

→ [[105, 166], [147, 206]],

judge: meaning preserved

1.000

identity_control

nothing ($X \rightarrow X$)

(byte-identical)

judge: meaning preserved

1.000